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WoodTapper: a Python package for explaining decision tree ensembles

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Introduction

Interpretable machine learning has become an increasingly critical concern (Nussberger et al., 2022; Sokol & Flach, 2024) as predictive models are deployed in high-stakes settings such as healthcare (Khalilia et al., 2011), marketing (Nguyen & Duong, 2021) or finance (Hassan & Abraham, 2016; Sakho et al., 2025) which is moreover a regulated sector. While complex models, such as tree-based ensemble methods, often yield strong predictive performance, their opacity can pose challenges for accountability, trust and compliance. Among interpretable models, rule-based methods are especially attractive because they are explained in the form of "if-then" statements, which are often easier to audit and communicate than latent feature transformations.

Statement of need

The original SIRUS algorithm (Bénard et al., 2021a, 2021b) offered a principled approach to generate simple and stable rule-based models from random forests. However, its implementations have been limited to R and Julia (Bénard et al., 2021b; Huijzer et al., 2023), creating accessibility barriers for the Python data science community. `WoodTapper` addresses this gap by offering a native Python implementation that integrates with the scikit-learn ecosystem. Furthermore, `WoodTapper` extends rules extraction (i) from all the tree-based models in scikit learn (Random Forest, Gradient Boosting and Extremely Randomized Trees) and (ii) to the multiclass classification setting.

In addition, `WoodTapper` introduces an example-based explainability methodology that can be applied to all scikit-learn tree-based models. This approach associates predicted samples with representative samples from the training data set, explaining tree-based models predictions through examples.

Rules Extraction Module

Formulation

In this section, we present our `RulesExtraction` module and we specifically consider its application to a random forest classifier, which corresponds to the SIRUS algorithm introduced by Bénard et al. (2021b).

We suppose that we have a training set $\mathcal{D}_n = \{(x_i, y_i)\}_{i=1}^n$ composed of n pairs taking values in $\mathbb{R}^p$ and $\{0, 1\}$ respectively (binary classification). We denote by $x_i^{(j)}$ the j-th component of

the i-th sample in $\mathcal{D}_n$. We suppose we have a set of trees $\{\mathcal{T}_m, m = 1, \dots, M\}$ from a tree ensemble procedure, each grown with randomness Θ_m.

In a tree $\mathcal{T}_m$, we denote by $\mathcal{P}$ a path of successive splits from the root node. $\mathcal{P}$ defines thus a hyperrectangle in the input space, $\hat{H}(\mathcal{P}) \subset \mathbb{R}^p$. We associate $\mathcal{P}$ with a rule function $\hat{g}_{\mathcal{P}}$ returning the mean of Y from the training sample inside and outside $\hat{H}(\mathcal{P})$.

For a set of trees $\{\mathcal{T}_m, m = 1, \dots, M\}$ and a path $\mathcal{P}$, we define:

$$\hat{p}\left(\mathcal{P}\right) = \frac{1}{M} \sum_{m=1}^{M} \mathbb{1}_{\{\mathcal{P} \in \mathcal{T}(\Theta_m, \mathcal{D}_n)\}},$$

which corresponds to the empirical probability that the path $\mathcal{P}$ belongs to the set of trees $\{\mathcal{T}_m, m = 1, \dots, M\}$. The set of final rules is $\{\hat{g}_{\mathcal{P}}, \mathcal{P} \in \hat{\mathcal{P}}_{p_0}\}$ where $\hat{\mathcal{P}}_{p_0} = \{\mathcal{P}, \hat{p}(\mathcal{P}) > p_0\}$ with $p_0 \in [0, 1)$. The final rules are aggregated as follows for building the final estimator:

$$\hat{\eta}_{p_0}(x) = \frac{1}{|\hat{\mathcal{P}}_{p_0}|} \sum_{\mathcal{P} \in \hat{\mathcal{P}}_{p_0}} \hat{g}_{\mathcal{P}}(x).$$

Beyond the binary classification detailed here, we also implemented the rule extractor for regression, where final rules are aggregated using weights learned via ridge regression.

Implementation and running time

`RulesExtraction` module adheres to the scikit-learn (Pedregosa et al., 2011) estimator interface and enables smooth integration with existing workflows involving pipelines, cross-validation, and model selection (see Table 1).

Table 1: Comparison of SIRUS implementations across softwares.

Feature	`WoodTapper` (Py)	SIRUS (R)	SIRUS (Jl)
Language	Python 3.x	R 4.x	Julia 1.x
Forest	`scikit-learn`	`ranger`	Own
Availability	PyPI (`woodtapper`)	CRAN (`sirus`)	General registry
Parallelism	✓ (via `joblib`)	Limited (via `parallel`)	✓ (native)
ML pipelines	✓	Partial	Partial
Tree models	All	random forest	random forest
Rules interface	Unified class methods	Function-based	Function-based
Tree depth ≥ 3	✓	✓	×
Classification	Multiclass	Binary	Multiclass

We compare the runtimes of SIRUS in Python (ours), R, and Julia using 5 threads on an AMD Ryzen Threadripper PRO 5955WX (16 cores, 4GHz) with 250GB RAM. We also experimented on large-scale industrial data sets, including from the banking sector, and observed the same trends as displayed here. SIRUS.jl exhibits higher runtime compared to Python and R implementations. The R version, relying on ranger, is faster for tree construction on large datasets than scikit-learn. Our Python implementation, however, is considerably more efficient for rule extraction, regardless of sample size or feature dimensionality (see Figures 1 and 2).

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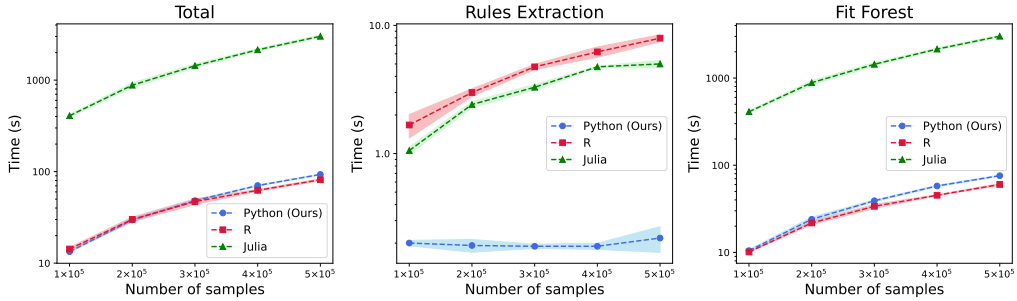

Figure 1: SIRUS running time for simulated data using 5 threads, with d=200 and M=1000.

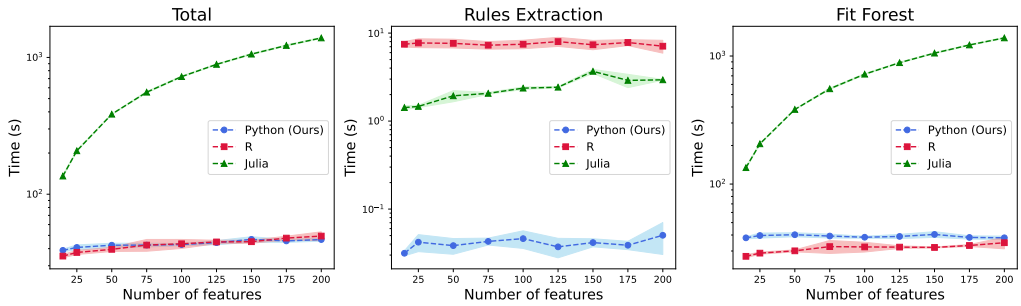

Figure 2: SIRUS running time for simulated data using 5 threads, with n=300K and M=1000.

Extracted rules and predictive performances

The rules produced by the original SIRUS (R) and our `RulesExtraction` in 3 and 4 on the Titanic data set are identical, and predictive performances in Table 2 are very similar, confirming that our implementation faithfully reproduces the original algorithm.

```
 [1] "Proportion of class 1 = 0.386 - Sample size n = 887"
 [2] "if Sex in {male} then 0.19 (n=573) else 0.742 (n=314)"
 [3] "if Pclass < 3 & Sex in {female} then 0.947 (n=170) else 0.252 (n=717)"
 [4] "if Sex in {male} & Age < 14.8 then 0.535 (n=43) else 0.378 (n=844)"
 [5] "if Pclass < 2 & Sex in {male} then 0.369 (n=122) else 0.388 (n=765)"
 [6] "if Sex in {male} & Fare < 14.5 then 0.117 (n=332) else 0.546 (n=555)"
 [7] "if Sex in {male} & Fare < 22.2 then 0.132 (n=380) else 0.576 (n=507)"
 [8] "if Sex in {male} & Fare < 10.5 then 0.107 (n=270) else 0.507 (n=617)"
 [9] "if Sex in {male} & Fare < 27.7 then 0.151 (n=430) else 0.606 (n=457)"
[10] "if Sex in {male} & Fare < 39.7 then 0.169 (n=485) else 0.647 (n=402)"
[11] "if Sex in {male} & Fare < 78 then 0.179 (n=542) else 0.71 (n=345)"
[12] "if Sex in {female} & Siblings.Spouses.Aboard < 3 then 0.775 (n=293) el
se 0.194 (n=594)"
```

Figure 3: SIRUS (R) rules on Titanic data set.

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```
  Condition                          THEN P(C1)      ELSE P(C1)
-----------------------------------------------------------------------
if   Sex is male                     then 19%                else 74%
if   Sex is female & Pclass <= 2.00  then 95%                else 25%
if   Sex is male & Age <= 14.80      then 53%                else 38%
if   Sex is male & Pclass <= 1.00    then 37%                else 39%
if   Sex is male & Fare <= 14.45     then 12%                else 55%
if   Sex is male & Fare <= 22.22     then 13%                else 58%
if   Sex is male & Fare <= 10.50     then 11%                else 52%
if   Sex is male & Fare <= 27.72     then 15%                else 61%
if   Sex is male & Fare <= 77.96     then 18%                else 71%
if   Sex is male & Fare <= 39.69     then 17%                else 65%
if   Sex is male & Fare <= 8.05      then 10%                else 48%
if   Sex is male & Pclass <= 2.00    then 27%                else 43%
```

Figure 4: WoodTapper SIRUS (Ours) rules on Titanic data set.

Table 2: Performance metrics for Titanic and House Sales datasets.

Dataset	Metric	SIRUS (original R)	Ours
Titanic	Accuracy	0.79 ± 0.03	0.78 ± 0.02
	(ROC) AUC	0.84 ± 0.04	0.84 ± 0.04
House Sales	MSE	0.35 ± 0.02	0.34 ± 0.01
	MAE	0.26 ± 0.01	0.26 ± 0.01

Example-based explainability module

Formulation

The `ExampleExplanation` module of `WoodTapper` is independent of the `RulesExtraction` module and provides an example-based explainability. It enables tree-based models to identify the most similar training samples to x, using the similarity measure induced by generalized random forests (Athey et al., 2019; Breiman, 2001). For a new sample x with unknown label and a decision tree $\mathcal{T}_m$, let $\mathcal{L}_m(x)$ denote the set of training samples that share the same leaf as x. We define the similarity $w(x, x_i)$ between x and x_i as:

$$w(x, x_i) = \frac{1}{M} \sum_{m=1}^{M} \frac{\mathbb{1}_{\{x_i \in \mathcal{L}_m(x)\}}}{|\mathcal{L}_m(x)|}.$$

Finally, the l training samples with the highest $w(x, x_i)$ values, along with their target values y_i, are shown as the examples that best explain the prediction of x by the tree-based ensemble model.

In python, the *skgrf* (Flynn, 2021) package is an interface for using the R implementation of generalized random forest, focusing on classifiers for specifics learning tasks (causal inference, quantile regression,...). For each task, the user can compute the kernel weights, equivalently to our leaf frequency match introduce above. Thus, we compare the kernel weights computation by *skgrf* and our module. We stress on the fact that our `ExampleExplanation` is designed for usual tree-based models such as random forest of extra trees and not specifically in a context of causal inference or quantile regression. In particular, the tree building of our forest is different from the one in *skgrf*.

Implementation and running time

Our Python implementation of `ExampleExplanation` adheres to the scikit-learn interface and is agnostic to the underlying tree ensemble (3). The standard `fit` and `predict` methods

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remain unchanged, while an additional `explanation` method provides example-based explanations for new samples. The user can also load an already trained tree-based model into an `ExampleExplanation` classifier.

Table 3: Comparison of GRF weight computations in several Python packages.

Feature	`WoodTapper` (Py)	skgrf (Py)
Forest implementation	`scikit-learn`	`ranger`
Language	Python	Python & R
Package availability	PyPI (`woodtapper`)	PyPI (`skgrf`)
scikit-learn API compatible	✓	✓
Tree-based models	All	Tree and random forest
GRF	×	✓

In figure 5, we compare the kernel weight computation runtime of `ExampleExplanation` and *skgrf* (Flynn, 2021) using the same hardware as previous experiments. `ExampleExplanation` is consistently faster.

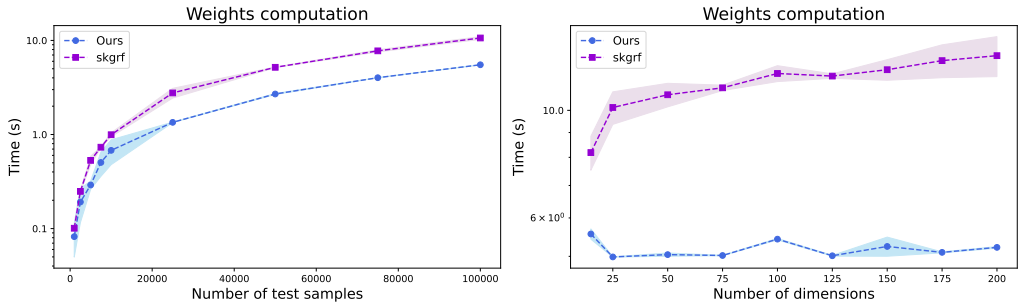

Figure 5: Weights computation running time for simulated data using.

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